

HDL - (Hardware Definition Language)

VHDL

```
-- Simple Logic Gates
architecture Behavioral of logic_demo is
begin
    and_out <= a and b;  -- Direct AND gate wiring
    or_out  <= a or b;   -- Direct OR gate wiring
end architecture;
```

Verilog

```
// Simple Logic Gates
module logic_demo (
    input  logic a, b,
    output logic and_out, or_out
);
    assign and_out = a & b;  // Direct AND gate
    assign or_out  = a | b;  // Direct OR gate
endmodule
```

HSL - (High-Level Sythesis)

C / C++ Frameworks

- AMD Xilinx Vitis HLS (Vivado HLS)
- Catapult HLS (Siemens)
- Intel HLS Compiler

Python Frameworks

- hls4ml
- PyMTL3 / PyRTL
- Migen / LiteX

Rust Frameworks

- Spade
- Veryl
- Rust-HDLs